Batted Ball Control

Does the hitter steer the ball, or does the pitch?

Trent Reimers | October 09, 2026

The question. Is a hitter's batted-ball distribution (pull angle and launch angle) set by his **swing shape** or by his **approach**, and which hitters *control* where the ball goes versus *react* to the pitch?

The answer. Mostly swing. Across the targets that can be explained, swing shape alone explains .41 of pull lean and .29 of pull sensitivity (cross-validated R^2), versus .19 and .14 for approach. Next season, swing shape still predicts .33 of pull lean; approach falls to .13. The "how" is swing direction, contact depth, and **timing to speed**: reactors let pitch speed decide where they meet the ball.

The spectrum. Two stable traits: **sensitivity** (how much the pitch moves his batted ball; year-to-year $r = .63$) and **pull lean** (how far he pulls beyond what the pitch implies; $r = .76$). They are linked (r = -.37 season by season, -.68 once seasons are pooled and noise drops out, across 482 hitters) but not the same thing.

PART 1

Why this question

What makes a direction gap repeat?

The direction-aware xBA project ("Another Hard-Hit Out") showed that batted-ball direction describes a season well, but a hitter's direction gap repeats only partly from one year to the next ($r \approx .35$). That leaves an open question: when a hitter's spray pattern *does* repeat, what is repeating? Two candidate answers:

- **Swing shape.** Bat speed, attack angle, swing path tilt, and timing put the ball in a characteristic place regardless of the pitch.
- **Approach.** The hitter chooses which pitches to swing at; his distribution reflects his pitch diet more than his swing.

The goal is a **spectrum**: at one end hitters who *control* their distribution (the ball goes where they want), at the other hitters who *react* ("just get to the pitch").

The design rule throughout: **control the pitch by its shape, never by its label.** Pitch-type calls are never model inputs. A cutter that moves like a slider is treated as a slider.

PART 2

Data and the pitch layer

Data

MLB Hawk-Eye pitch data, 2023–2026 (post-shift-ban seasons only). Hitter metrics use 1,337 hitter-seasons (560 hitters) with 150+ batted balls. Batted balls include home runs; bunts and sacrifices are dropped. Bat-tracking swing traits use 2024 onward only (2023 tracked about 42% of swings; 2024 93%; 2025–26 97%).

Shape-based pitch families

Families are built from explicit shape rules on pitcher-season-pitch averages, not k-means (clusters drifted between seasons, relabeling the same cutter as a riding fastball one year and a cutter the next). Two ideas carry most of the weight:

- **Movement relative to release.** Expected IVB and arm-side run for a primary fastball are modeled from arm angle and release height; *ride over expected* is actual minus expected. A 10" IVB fastball from a low slot can out-ride a 15" one from on top.
- **Flatness.** VAA over expected at the same plate height. A fastball is "riding" if it rides 2"+ over expected **or** is 0.5° flatter than expected.

Family assignment is stable: the same pitcher's same pitch keeps its family 87% of the time across consecutive seasons. The spot check below is where the rules earned their keep: Hader's Statcast "sinker" is a riding fastball by shape; Glasnow, Clase and Jansen throw cut *fastballs*, not cutters; Bibee's cutter groups with sliders; Strider's ride loss after surgery moves him from riding to average.

Table 1: Spot check: shape families vs. Statcast labels (latest season per pitch)

Pitcher	Season	Statcast	Family	Velo	IVB	Run (arm)	Ride OE	Flat (deg)
Emmanuel Clase	2025	FC	Cut fastball	98.9	12.5	-4.1	−3.5	−0.00
Freddy Peralta	2026	FF	Riding fastball	94.3	16.7	7.0	+2.6	+0.79
Josh Hader	2026	SI	Riding fastball	95.1	18.6	8.9	+6.1	+0.45
Kenley Jansen	2026	FC	Cut fastball	92.5	16.5	-5.0	−0.8	−0.50
Logan Webb	2026	FC	Cutter	89.8	6.9	1.0	−3.7	−0.08
Logan Webb	2026	FF	Average fastball	92.3	9.9	7.6	−0.7	+0.29
Logan Webb	2026	SI	Sinker / running fastball	92.1	0.1	15.5	−10.5	−0.59
Seth Lugo	2026	FC	Cutter	89.8	11.9	0.4	+0.1	−0.28
Seth Lugo	2026	FF	Riding fastball	91.7	15.3	8.9	+3.9	+0.28
Seth Lugo	2026	SI	Sinker / running fastball	91.4	7.8	16.4	−1.8	−0.19
Spencer Strider	2026	FF	Average fastball	95.1	17.4	8.9	+1.9	+0.33
Tanner Bibee	2026	FC	Slider	85.8	2.1	-6.9	−14.2	−2.18
Tanner Bibee	2026	FF	Average fastball	94.0	15.6	4.7	−1.7	−0.46
Tanner Bibee	2026	SI	Sinker / running fastball	93.2	9.1	14.6	−6.3	−0.94
Tyler Glasnow	2026	FF	Cut fastball	95.5	16.0	0.6	−1.2	+0.00
Tyler Glasnow	2026	SI	Sinker / running fastball	95.8	14.3	10.8	−2.8	−0.14

PART 3

The pitch-only league model

What the pitch alone predicts

An xgboost model predicts the pull angle and launch angle a league-average hitter would produce on each pitch, from the pitch alone: location (relative to the zone), velocity, full shape in the batter's frame, release, approach angles, shape relative to the pitcher's primary fastball, count, and handedness. Trained 2023–2025 with tree counts chosen on 2025; scored on 2026. Every batted ball gets a leave-one-season-out expectation.

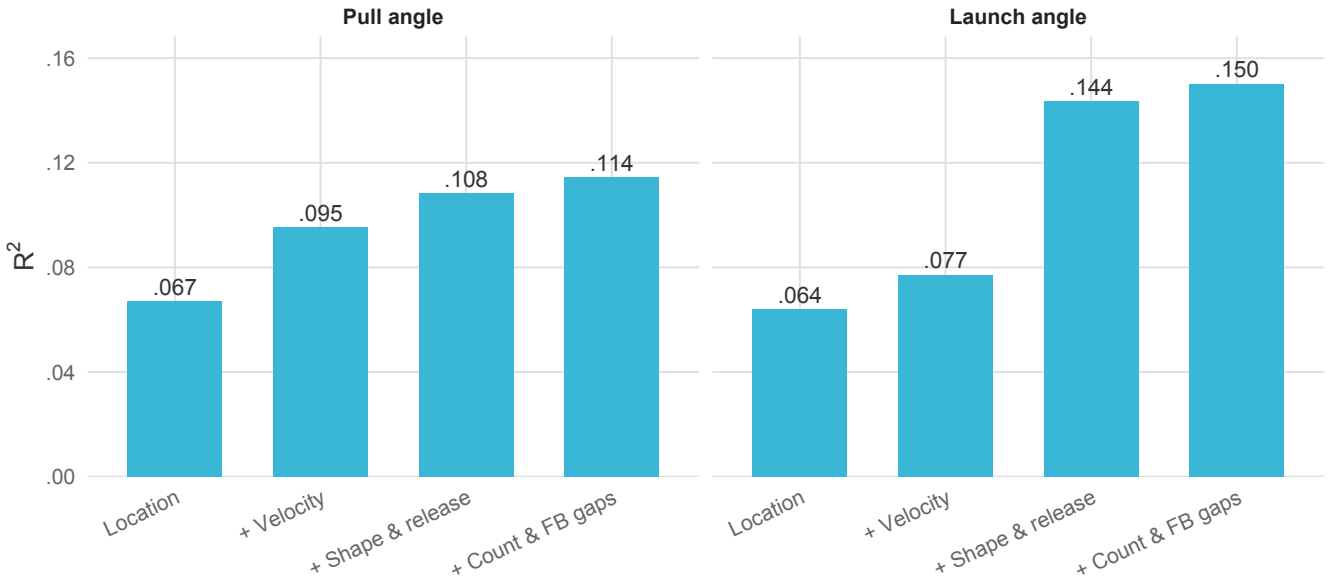


Figure 1: Variance in batted-ball outcome explained by the pitch, built up one block at a time (2026 test season).

Direction is location plus timing; launch angle is shape. For pull angle, velocity adds .028 and shape only .013; the top inputs are horizontal location (31%), velocity gap to his fastball (16%), vertical approach angle (15%). For launch angle, shape adds .066, more than double its contribution to direction; the top inputs are pitch height (16%), horizontal location (12%), ride over expected (11%). Count adds little to either.

The pitch explains only 11% of pull angle and 15% of launch angle. The other 85%–89% is hitter, swing, timing, and noise, which is the room this project lives in.

PART 4

Sensitivity and pull lean

Two numbers per hitter

For each hitter-season, a mixed model regresses actual on expected over his batted balls:

$$\text{actual}_i = (\beta_0 + u_{0h}) + (\beta_1 + u_{1h})\,\text{expected}_i + \varepsilon_i$$

- **Sensitivity** $= \beta_1 + u_{1h}$. Near 1: the ball goes where the pitch sends it (reactor). Near 0: he puts it where he wants regardless (controller). Above 1: he overreacts.
- **Pull lean (imprint)** $= u_{0h}$. A consistent lean beyond the pitch: "pulls 8° more than league on identical pitches."

Both are shrunk toward the league, and both are estimated separately for pull angle and launch angle.

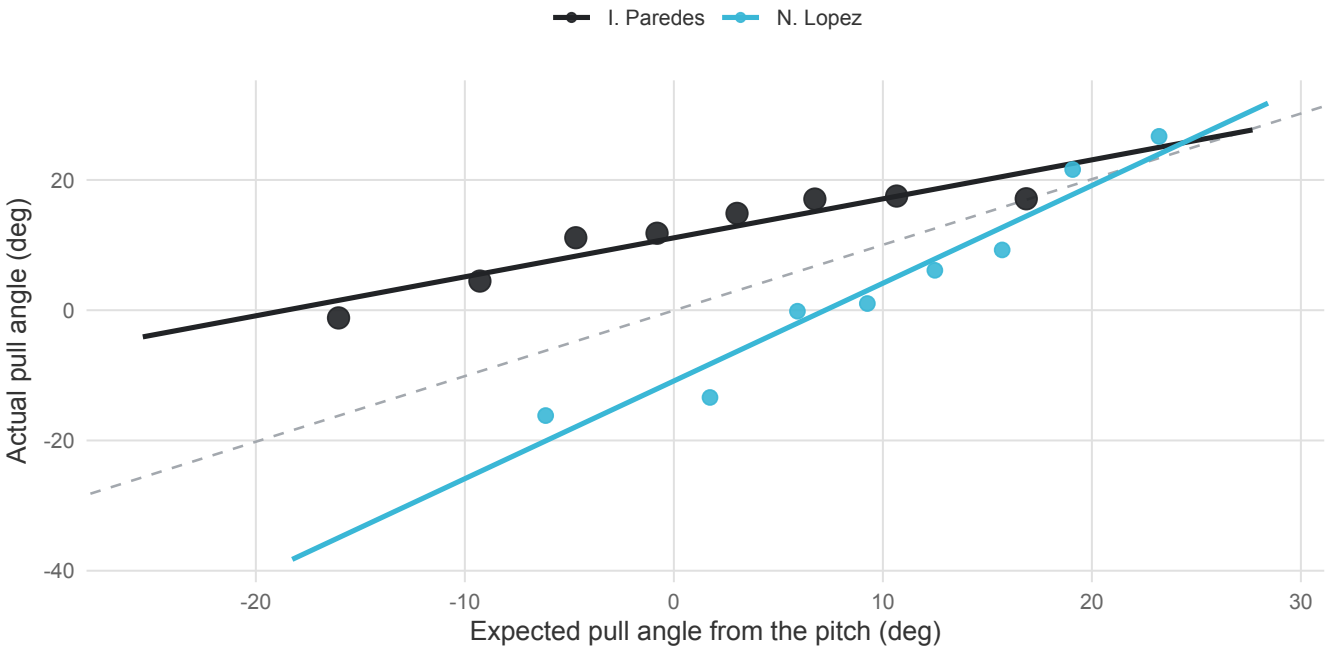


Figure 2: Same pitches, different hitters. Points: mean actual pull angle in octiles of expected pull angle. Dashed: league. I. Paredes barely moves with the pitch; N. Lopez goes where it takes him.

The raw slopes make the definitions concrete: I. Paredes 0.60 vs. N. Lopez 1.50 (league 1.01). The shrunken sensitivities used below are 0.69 and 1.31.

The traits are real and they repeat

League slopes of 1.05 (pull) and 1.03 (LA) mean the expectations are calibrated. Hitters differ in sensitivity by a true SD of .18 (pull) and .14 (LA); a likelihood-ratio test rejects "everyone has the same slope" at p $\approx 10^{-207}$.

Pull sensitivity (.63 year to year) and both leans (.76, .78) are stable traits. **LA sensitivity is the**

Table 2: Do hitters differ in sensitivity?

Target	League slope	True SD of sensitivity	LRT p (slope varies)
Pull angle	1.05	.18	1e-207
Launch angle	1.03	.14	1e-122
Pull, two strikes	1.03	.18	1e-73
LA, two strikes	1.01	.14	1e-26

Table 3: Reliability of hitter metrics

Metric	Split half	Year to year
Pull sensitivity	.77	.63
Pull lean	.87	.76
LA sensitivity	.53	.49
LA lean	.90	.78

weak one (.49), and the rest of the report treats it accordingly.

Two strikes. Hitters differ in two-strike sensitivity by about as much as in all counts (true SD .18 vs. .18), and a hitter's two-strike sensitivity tracks his full-count one (r = .86): protecting with two strikes does not reshuffle who controls and who reacts.

The foul-line check

Sensitivity and pull lean are correlated. One worry: pull-line hitters lose their most-pulled balls to foul territory, which could compress their slope mechanically. To test it, each hitter was simulated with his own pitches, his own lean, and sensitivity fixed at exactly 1, with balls outside ±45° dropped.

Table 4: Foul-line truncation check

Check	Value
r(simulated no-control slope, pull lean)	-.07
r(sensitivity, lean): observed vs corrected	-.37 vs -.33
Split-half reliability: observed vs corrected	.62 vs .55
Year-to-year: observed vs corrected	.53 vs .48

The simulated slopes barely depend on lean (r = -.07): truncation compresses everyone about equally, so it cannot manufacture differences between hitters. After correction the sensitivity–lean link is -.33 (was -.37) and reliability barely moves. The corrected values' absolute scale is off, so they are a check, not a metric; the conclusion stands that **sensitivity and lean are two real, partly linked traits**, which makes the spectrum two-dimensional.

PART 5

The spectrum

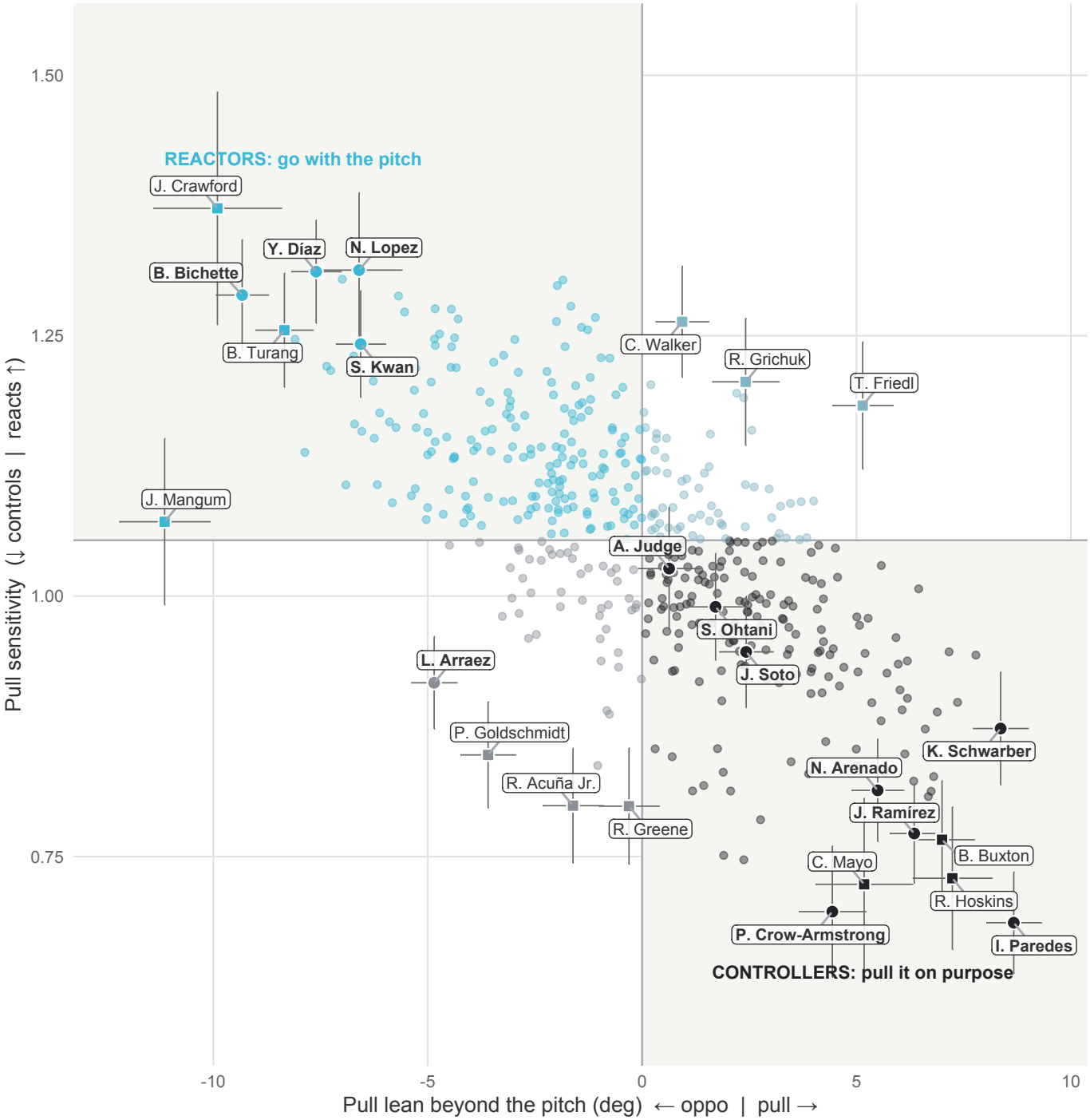


Figure 3: Batted ball control spectrum: 482 hitters with 300+ batted balls, precision-weighted across 2023–2026. Lines: zero lean and league slope (1.05). Circles: named archetypes; squares: most extreme other hitters per quadrant. Bars: ±1 SE.

Reading the chart: the **lower right** is hitters who pull more than the pitch implies *and* whose direction barely moves with the pitch (38% of hitters); the **upper left** is hitters who lean oppo and go where the pitch sends them (39%). The off-diagonal corners (23% combined) are why this is two dimensions, not one: even pooled, lean and sensitivity correlate at -.68, not −1 (and only -.37 within a single season).

Archetype check: 13 of 13 named hitters land where expected.

Table 5: Most controlled: lowest pull sensitivity

Hitter	Seasons	BBE	Sensitivity	Pull lean	Control pctl
Isaac Paredes	2023–26	1,611	0.69	+8.7	100
Pete Crow-Armstrong	2024–26	1,137	0.70	+4.4	100
Coby Mayo	2025–26	451	0.72	+5.2	100
Rhys Hoskins	2024–26	666	0.73	+7.2	99
Anthony Santander	2023–25	1,046	0.75	+2.4	99
Eddie Rosario	2023–24	580	0.75	+1.9	99
Byron Buxton	2023–26	1,073	0.77	+7.0	99
José Ramírez	2023–26	2,018	0.77	+6.3	99
Salvador Perez	2023–26	1,813	0.79	+2.8	98
Riley Greene	2023–26	1,384	0.80	−0.3	98
Ronald Acuña Jr.	2023–26	1,242	0.80	−1.6	98
Tony Kemp	2023	312	0.81	+6.7	98

Table 6: Most reactive: highest pull sensitivity

Hitter	Seasons	BBE	Sensitivity	Pull lean	Control pctl
Justin Crawford	2026	337	1.37	−9.9	0
Nicky Lopez	2023–26	703	1.31	−6.6	0
Yandy Díaz	2023–26	1,909	1.31	−7.6	1
Ke'Bryan Hayes	2023–26	1,264	1.30	−7.0	1
Isaac Collins	2025–26	601	1.30	−1.8	1
Kyle Isbel	2023–26	1,021	1.30	−2.0	1
Bo Bichette	2023–26	1,707	1.29	−9.3	1
Nathaniel Lowe	2023–26	1,490	1.29	−5.7	2
Troy Johnston	2025–26	397	1.28	−1.7	2
Jarren Duran	2023–26	1,584	1.28	−4.8	2
Joey Ortiz	2024–26	1,016	1.28	−4.4	2
Wander Franco	2023	377	1.27	−5.5	2

The full table (all 482 hitters, with SEs, quadrant, and a one-dimensional control percentile) is written to `spectrum_hitters.csv`.

PART 6

Swing shape or approach?

Part A: the original question

Two blocks of hitter-season predictors, used to explain sensitivity and lean:

- **Swing shape**: season averages over *all* tracked swings, including whiffs and fouls: bat speed, swing length, swing path tilt, attack angle.
- **Approach**: zone swing rate, chase rate, swing rate by shape family, and the location of swings.

Per-swing contact depth and attack direction are *responses* to the pitch, so they are never used as controls (the mediator trap). They appear only in Part B, as mechanism.

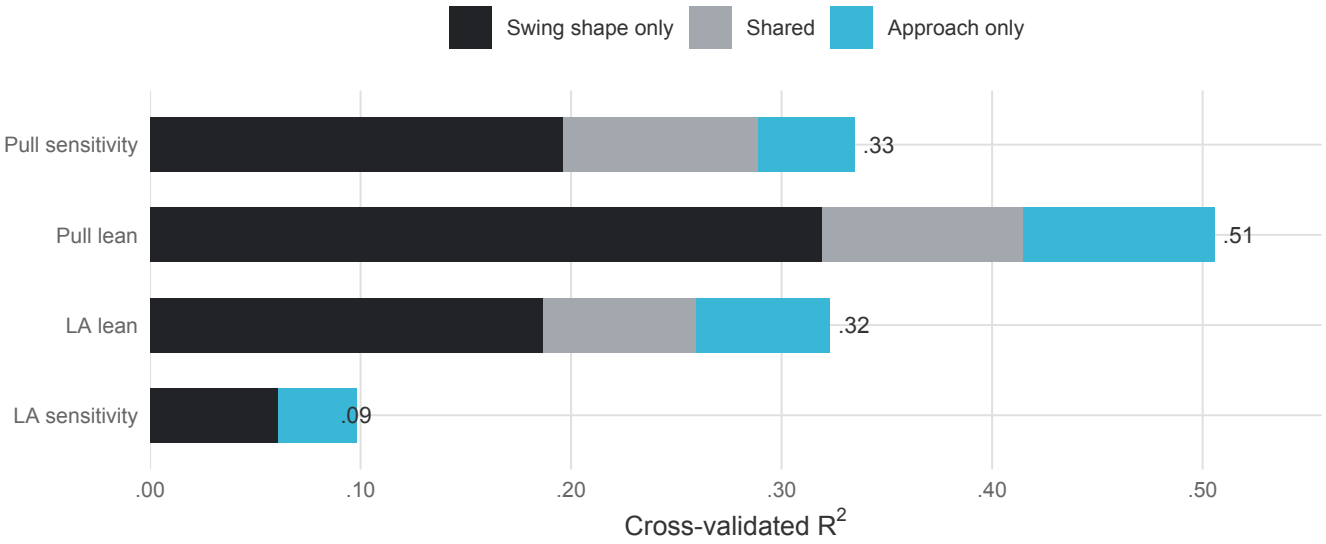


Figure 4: Commonality analysis: cross-validated R^2 split into what only swing shape explains, what only approach explains, and what they share.

Table 7: Same season (cross-validated) and next-season prediction

	Same season						Next season	
Target	**Swing**	**Approach**	**Both**	**Swing only**	**Approach only**	**Shared**	**Next: swing**	**Next: approach**
Pull sensitivity	.29	.14	.33	.20	.05	.09	.23	.09
Pull lean	.41	.19	.51	.32	.09	.10	.33	.13
LA lean	.26	.14	.32	.19	.06	.07	.23	.05
LA sensitivity	.05	.03	.09	.06	.04	-.01	.03	-.00

Swing shape wins every explainable trait. Approach explains about 0.5× as much on average, and most of what it explains is shared with swing shape. The next-season test is sharper: swing shape keeps about 83% of its same-season power, approach about 59%. Swing shape is the better predictor of next year on every trait. LA sensitivity is barely explainable by anything (.09 with both blocks), consistent with its weak reliability.

Attack angle dominates. One SD of attack angle is worth +2.1° of pull lean (t = 21) and lower pull sensitivity (t = -15).

Table 8: Strongest individual drivers (standardized)

Target	Block	Term	Per SD	t
LA lean	Swing Shape	Attack angle	+2.0	14.7
LA lean	Approach	Inside-ness of swings	−0.7	-6.2
LA lean	Approach	Zone swing %	+2.4	5.7
LA lean	Approach	First-pitch swing %	−1.2	-5.6
LA lean	Approach	Swing % vs breaking	−1.7	-4.9
LA lean	Approach	Chase %	+1.6	4.2
LA lean	Approach	Swing % vs fastballs	−1.3	-3.5
LA lean	Swing Shape	Swing length	+0.5	3.0
Pull lean	Swing Shape	Attack angle	+2.1	21.1
Pull lean	Approach	Inside-ness of swings	−1.2	-13.2
Pull lean	Swing Shape	Swing path tilt	−0.9	-9.5
Pull lean	Swing Shape	Swing length	+0.7	6.3
Pull lean	Swing Shape	Bat speed	−0.4	-3.3
Pull lean	Approach	Swing % vs breaking	−0.8	-3.3
LA sensitivity	Swing Shape	Swing path tilt	−0.025	-7.3
LA sensitivity	Approach	First-pitch swing %	−0.021	-4.0
LA sensitivity	Approach	Inside-ness of swings	+0.011	3.7
LA sensitivity	Approach	Chase %	−0.033	-3.4
LA sensitivity	Approach	Swing % vs breaking	+0.027	3.1
LA sensitivity	Swing Shape	Bat speed	+0.012	3.0
Pull sensitivity	Swing Shape	Attack angle	−0.062	-14.7
Pull sensitivity	Approach	Inside-ness of swings	+0.029	7.7
Pull sensitivity	Swing Shape	Swing length	−0.019	-3.8

Approach's real contribution is *where* he swings. Hitters who swing at more inside pitches are more sensitive and show less pull lean *beyond expected*: they pull by pitch choice rather than by swing. That is a form of control the contact-only metrics cannot see, which is why the approach block exists.

Is attack angle partly timing? Measured attack angle rises with earlier contact, so some of its power could be timing in disguise. Removing the part of attack angle explained by average contact depth:

Table 9: Attack-angle timing check (cross-validated R^2)

Target	Swing (raw)	Swing (timing removed)	Approach	Margin
Pull sensitivity	.29	.23	.14	.09
Pull lean	.42	.31	.19	.12
LA sensitivity	.05	.05	.03	.03
LA lean	.26	.18	.14	.04

Swing shape still wins every explainable trait; about a quarter of attack angle's power was timing, and the narrowest margin is LA lean (.04).

PART 7

How it happens

Part B: the mechanism

Part B uses the per-swing responses that Part A had to exclude: how a hitter's swing direction and contact depth shift with the pitch. It is reported separately because it is partly circular (these are pieces of the outcome), but it answers *how* control works.

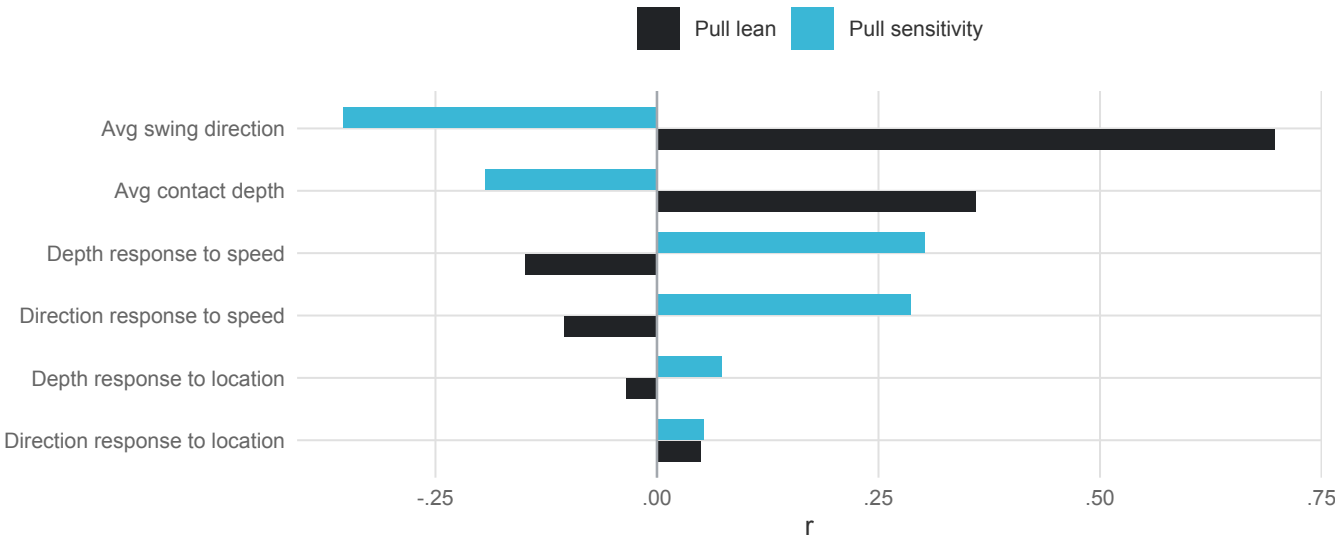


Figure 5: Correlation of each swing behavior with the hitter trait.

Pull lean is swing direction plus contact depth. Average swing direction ($r = .70$) and average depth ($r = .36$) lead the mechanism measures, which together explain .54 of it: a pull hitter aims pull-side and meets the ball out front.

Sensitivity is not about adjusting to location. The natural guess, that controllers keep the same swing direction regardless of location, fails: direction response to location correlates .05 with sensitivity. The missing piece is **timing to speed**. Adding how contact depth and direction shift with pitch speed raises the mechanism R^2 for sensitivity from .14 to .25 ($r = .30$ and .29). Reactors let the pitch's speed decide where they meet it.

Table 10: Timing, location response, and where they aim

Hitter	Seasons	Depth per mph (in)	Direction response	Avg aim (deg)	Avg depth (in)	Sensitivity
Bo Bichette	2024–26	1.53–1.73	−3 to −1	−3.5	23.6	1.29
Steven Kwan	2024–26	0.74–0.81	+11 to +13	+0.3	27.6	1.24
Nolan Arenado	2024–26	0.88–1.04	+3 to +5	+8.4	34.4	0.81
José Ramírez	2024–26	0.74–0.83	+13 to +15	+7.8	31.7	0.77
Isaac Paredes	2024–26	0.82–0.95	+5 to +7	+13.0	36.2	0.69

Timing is one clear route to reacting. Bichette meets slower pitches 1.53–1.73 in farther out front per mph (league median 1.20); his batted ball follows the pitch's speed.

Where a hitter aims is the other big piece. Kwan and Ramírez look alike on both responses: low timing response (0.74–0.81 vs. 0.74–0.83) and a swing direction that follows location (+11 to +13 vs. +13 to +15). Yet Kwan is one of the most reactive hitters and Ramírez one of the most controlled. What separates them is where they aim: Ramírez's swing points +7.8 degrees pull-side and meets the ball 31.7 in out front on average; Kwan's points +0.3 and meets it at 27.6 in. A hitter who aims pull-side and out front keeps pulling whatever the pitch does; one who aims up the middle leaves room for the pitch to move the ball both ways.

The mechanism measures explain about 25% of sensitivity, so a good part of *how* control works is still outside what bat tracking measures today.

PART 8

Limitations

What this does not show

- **LA sensitivity is weak and unreliable** (year to year .49, split half .53) and is barely explainable. The launch-angle story here rests on LA *lean*.
- **Bat tracking starts in 2024.** Swing traits cover three seasons; next-season tests use 485 back-to-back hitter pairs.
- **Only contact is observed.** Sensitivity and lean describe the balls a hitter put in play. Refusing to swing at certain pitches is also control; the approach block captures some of it, but the contact metrics cannot.
- **Shrinkage.** Hitter metrics are shrunk toward the league; hitters with few batted balls look more average than they are.
- **Mechanism is partly circular.** Part B variables are components of the outcome; they explain *how*, not *why*.
- **Foul-line correction is a check, not a metric.** Its absolute scale is off; only its effect on correlations and reliability is used.

PART 9

Reusable outputs

Files

All under `/Volumes/X9 Pro/Baseball Projects/Code Repo/batted_ball_control/`.

File	Contents	Use
pitch_layer_{year}.parquet	Per-pitch shape in batter and pitcher frames, ride/run over expected, flatness, fastball gaps, family	Any shape-controlled pitch analysis
pitch_families.csv / pitch_family_groups.csv	Rule-based shape families by pitcher-season-pitch	Reporting and pitch deep dives
expected_bbe.parquet	Leave-one-season-out expected pull and launch angle for every batted ball	Pitch-adjusted spray for any hitter question
hitter_control.csv	Sensitivity and lean (pull and LA, plus two-strike) with SEs, by hitter-season	Scouting: who steers vs. who reacts
control_tests.csv / control_reliability.csv	Mixed-model tests and reliability	Reference
spectrum_hitters.csv	Pooled spectrum: sensitivity, lean, quadrant, control percentile	Spectrum chart, lookups